




TAQA Arabia

Agriculture Clients

Integrated Energy & Utility Solutions · Jan 2026

A World of Energy, Delivered End-to-End

- Founded in 2006 and listed on the EGX since 2023, TAQA Arabia is Egypt's largest private-sector energy and utility developer.
- Across four divisions — Gas, Power, Petroleum and Water — TAQA finances, builds, owns and operates the utility backbone of residential communities, industrial zones and touristic destinations.
- For a residential developer, that means one accredited partner can deliver gas, electricity, water, back-up power and EV charging under a single relationship.

Active member of the International Gas Union (IGU)

Accredited by the IGEM (Institution of Gas Engineers & Managers)

Gas

- Gas distribution & EPC networks
- NGV stations & car conversion
- Mobile CNG & LNG (virtual pipeline)

Power

- Generation & distribution (+1,600 MVA)
- Renewable energy & solar PV
- EV charging stations

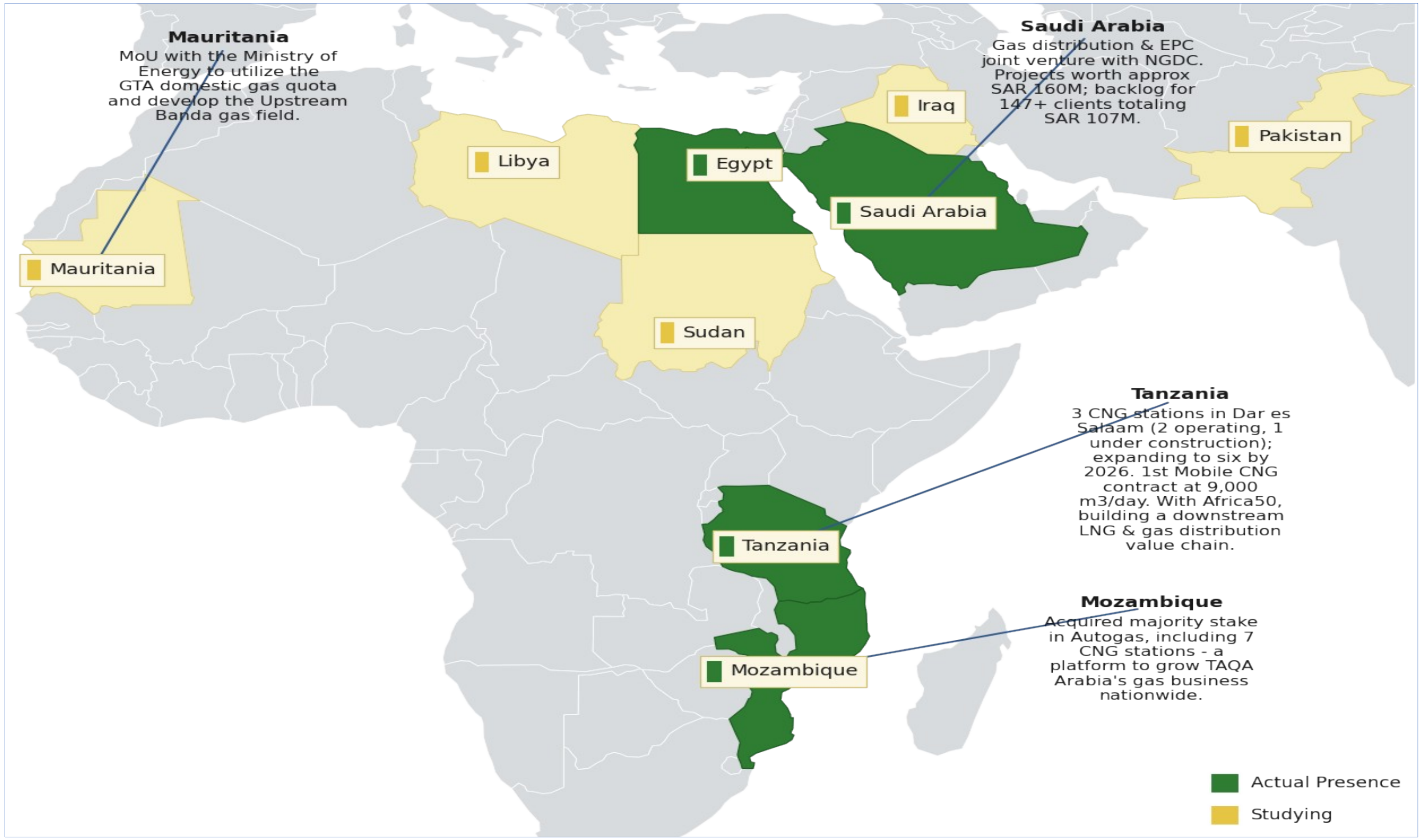
Petroleum

- Oil-marketing stations & terminals
- Lubricants retail & distribution
- Bulk fuel storage & logistics

Water

- Reverse-osmosis desalination
- Filtration & chemical treatment
- Smart, solar-powered operations

Presence Across Africa & the Middle East



Actual presence and markets under study across Africa & the Middle East — extending TAQA's integrated energy model well beyond Egypt.

Reach & Scale

8
Countries of presence
Egypt, GCC, Africa

4
Operating divisions
Gas · Power · Petroleum · Water

20+
Governorates in Egypt
Industrial, residential & touristic

3,400+
Employees
Across all divisions

TAQA Arabia · In Numbers

THE GROUP AT A GLANCE

EGP 13.4bn
Revenue — FY 2025

EGP 1.5bn
EBITDA — FY 2025

~6.5M
Residential gas customers

Founded 2006 · Listed on EGX 2023 · **3,400+** employees across all divisions

Operating Scale Across the Divisions

GAS

+10,000
km network

8
governorate concessions (15-yr)

POWER

+1,600
MVA distribution

+150
MW generation capacity

WATER

+47,000
m³/day desalination

15
operational locations

MOBILITY & CNG

86
CNG stations

1st
private EV-charging license

The Challenge Facing Egypt's Farms

Egypt is reclaiming millions of feddans of desert farmland — almost all of it off-grid.

Expensive Diesel Dependence

Pivots and pumps run on diesel gensets — the single largest and most volatile cost on the farm, exposed to every fuel-price spike.

No Pipeline, No Grid

Remote reclamation land sits far from the gas pipeline and the electricity grid, with no easy connection in sight.

Water Insecurity

Over-abstracted groundwater and unreliable supply put crop yield — and the whole investment — at risk.

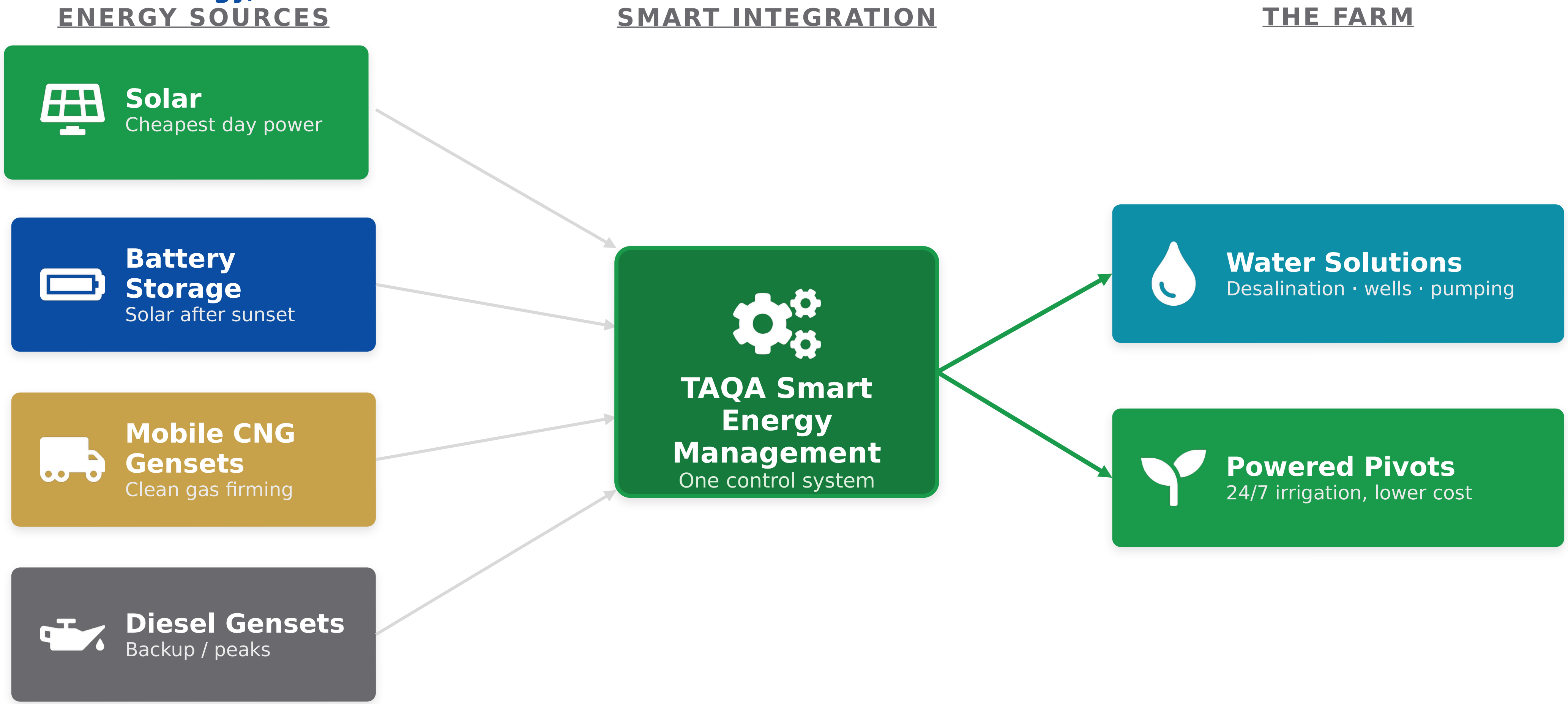
Fragile, Unmanaged Power

Stand-alone gensets fail, waste fuel at part-load and offer no backup — an interruption can cost an entire irrigation cycle.

The TAQA Integrated Farm Solution

One partner connects water and a smart, multi-source power stack to keep every pivot turning

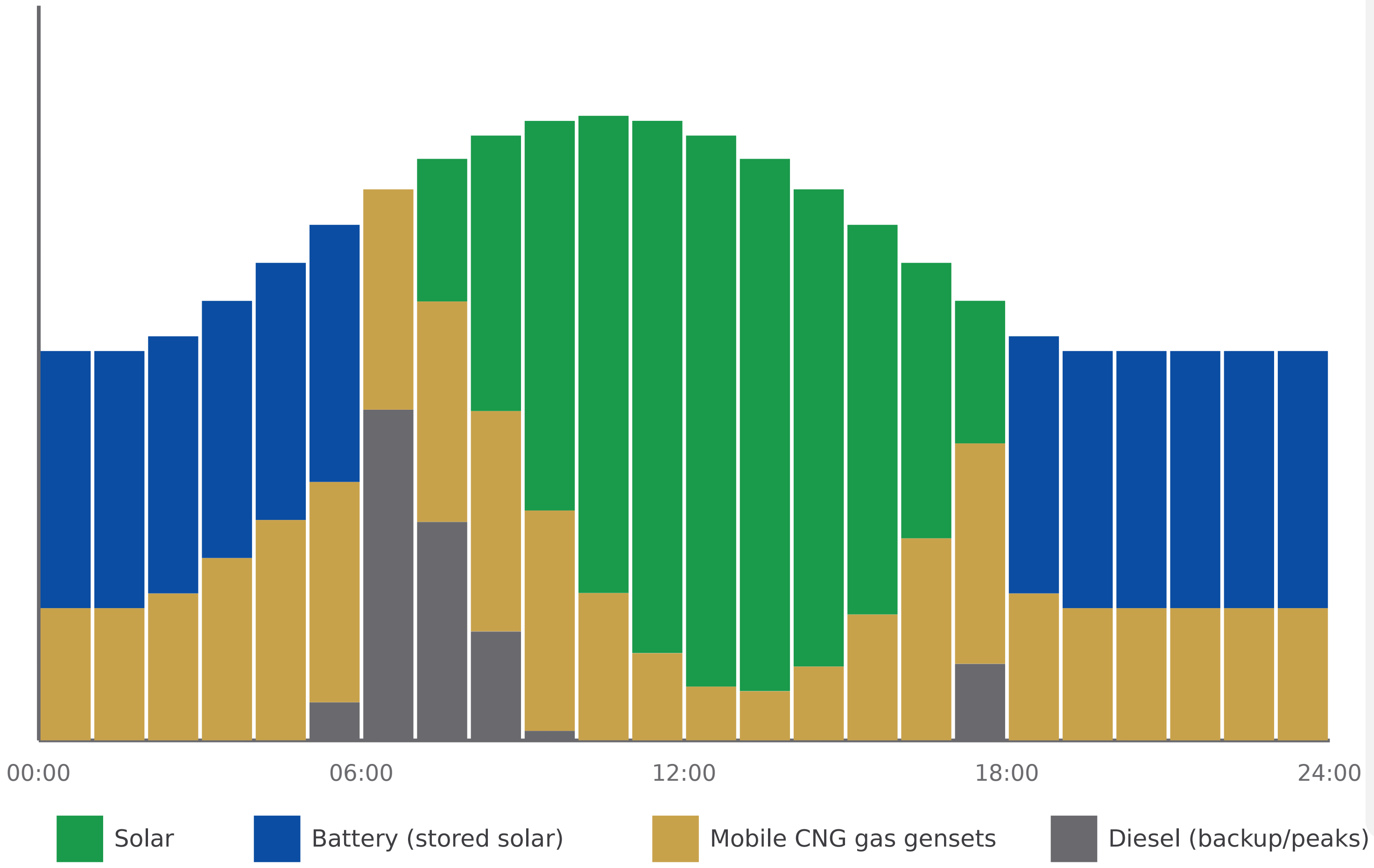
TAQA delivers the whole farm energy-and-water backbone as one system — **we secure the water, then power the pivots with the cheapest, cleanest energy, around the clock.**



The Smart Energy Mix: Power Around the Clock

Solar by day, batteries after sunset, gas and diesel for firming — least-cost, lowest-carbon, always on

Farm power demand (pivots + pumps)



How the stack is dispatched

1 Solar first
Cheapest, cleanest power runs the pivots all day.

2 Then batteries
Stored daytime solar carries irrigation into the evening and early morning.

3 Gas gensets firm
Mobile-CNG gensets fill gaps at ~40% less than diesel.

4 Diesel last
Runs only for peaks and backup — minimised, not relied on.

One TAQA energy-management system optimises the mix every minute — least cost, lowest carbon, always on.

Agriculture · Solutions Overview

Water Solutions 01

Desalination, groundwater treatment and smart, solar-powered irrigation pumping.

Solar 02

Tailored solar PV via PPA to power pivots and pumps by day — no farm capex.

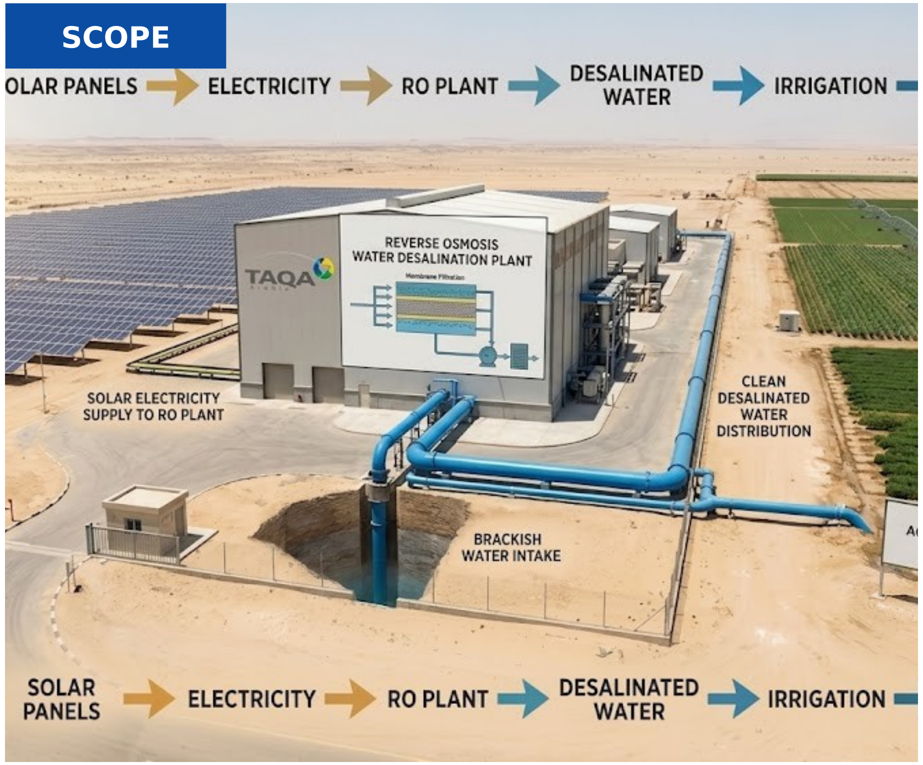
Mobile CNG 04

Portable natural gas to fuel farm gensets — ~40% cheaper than diesel.

Battery Storage 03

Shifts solar into the night and firms the stack so pivots never stop.

1 Water Solutions: Scope & How It Works



TAQA funds, builds, owns & operates the asset.
You pay only for the water you use — little to no upfront CapEx.

1 TAQA INVESTS THE CAPEX

Brackish / seawater RO desalination units · Pre-treatment & filtration system · Storage reservoirs & pumping · Irrigation distribution & smart meters · BOO / BOOT plant financing

2 THE PROCESS — STEP BY STEP

1. TAQA studies the farm's water source, salinity and demand.
2. Brackish or seawater is drawn in and pre-treated.
3. Reverse-osmosis units desalinate water to irrigation quality.
4. Treated water is stored, pumped and distributed to fields.
5. TAQA operates the plant and guarantees volume and quality.

3 WHAT YOU RECEIVE

- Reliable irrigation-grade water on demand
- Independence from strained groundwater
- Guaranteed volume, quality and uptime

Water for every acre

Water Solutions: Value Proposition

VALUE PROPOSITION

WHAT YOU GAIN

Guaranteed Water Security

Reliable freshwater for irrigation, independent of grid-utility and groundwater constraints.

Lower Energy & Water Cost

VSDs, peak-demand control and solar-powered pumping cut energy use and non-revenue water losses.

Frees Up Groundwater

Desalination and treatment reduce over-abstraction, protecting the aquifer for the long term.

Sustainability & ESG Impact

Solar-powered desalination, brine management and a reduced freshwater-extraction footprint.

THE TAQA ARABIA EDGE

Flexible Delivery Approach

BOO / BOOT or EPC via Capacity-as-a-Service — no upfront infrastructure cost for the community developer.

Smart Operations & Uptime

Real-time monitoring, leak & failure detection and predictive-maintenance dashboards.

Tailored to the Farm

Solutions sized to crop type, hectares and irrigation schedule — not one-size-fits-all.

Performance Guarantees

Volume and quality backed by contractual SLAs.

Water Solutions: Implementation Timeline

IMPLEMENTATION TIMELINE — START TO FINISH

Day 0–20	Day 20–60	Day 45–75	Day 75–210	Day 210–260	Day 260+
1	2	3	4	5	6
Water Demand Study	**Plant Design**	**Commercial Model**	**Construction**	**Commissioning**	**Smart O&M**
Crop water need & source-water survey.	RO sizing, pumping & solar integration.	BOO/BOOT & water-purchase agreement.	Plant, wells, networks & smart metering.	Testing, water-quality validation & handover.	IoT monitoring & predictive maintenance.

STUDY & DESIGN *Day 0–60*
Crop water-demand profiling, source-water survey, RO and pumping sizing, and solar-integration planning for the farm.

FINANCE & BUILD *Day 45–260*
BOO/BOOT structuring and water-purchase agreement, plant and well construction, irrigation networks, smart metering and water-quality validation.

SMART OPERATE *Day 260+*
IoT monitoring of water quality and flow, leak detection, predictive maintenance and energy-optimised pumping.

Water Solutions: Proven Track Record

Eco Solar Desalination — Soma Bay

Egypt's first & largest eco solar-powered desalination plant

On the Red Sea, TAQA Water built Egypt's first green desalination facility powered entirely by renewable energy, using technology that consumes 50% less power than peers. The same solar-powered, smart-operated water model — proven at scale — secures irrigation supply for large farms and agribusiness.

+47,000
m³/day contracted desalination

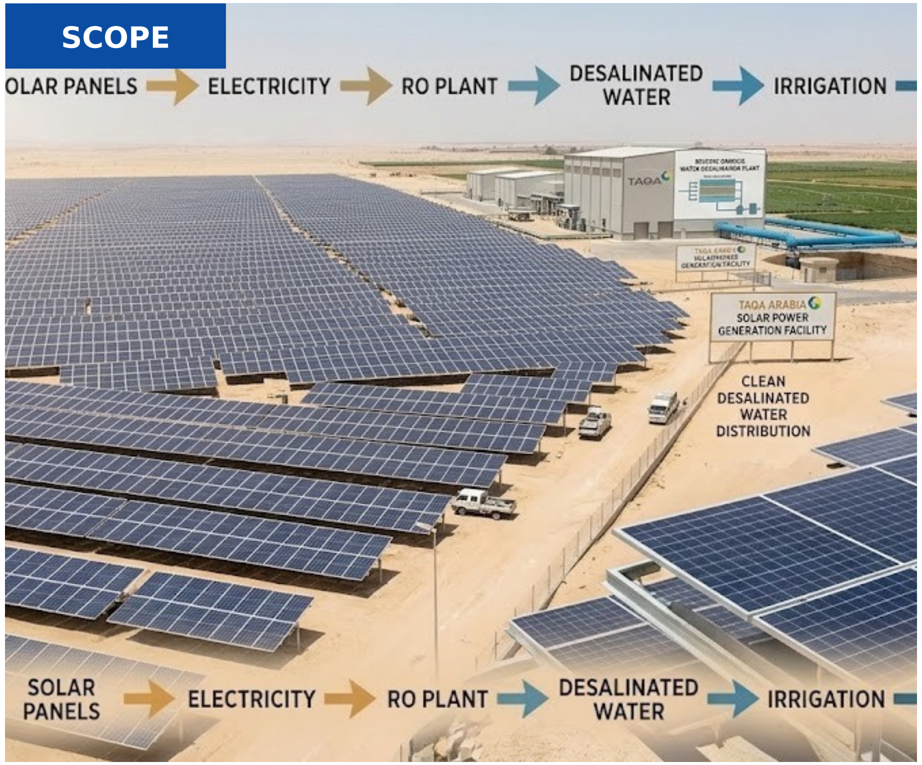
15
Operational water locations

50%
Less power than peer plants

8,560
Tonnes CO_2 avoided / year

2 Solar: Scope & How It Works

SCOPE



TAQA funds, builds, owns & operates the asset.
You pay only for the solar power you use — little to no upfront CapEx.

1 TAQA INVESTS THE CAPEX

Ground-mount & pump-side PV arrays · Solar inverters & mounting · Pump controllers & soft starters · Monitoring & connection works · BOO / BOOT solar financing

2 THE PROCESS — STEP BY STEP

1. TAQA assesses land, sun hours and the farm's pumping load.
2. PV arrays and inverters are installed near pumps and facilities.
3. Panels generate clean power through daylight irrigation hours.
4. Solar drives pumps, cold rooms and farm loads directly.
5. TAQA monitors yield and maintains the system.

3 WHAT YOU RECEIVE

- Daytime power matched to irrigation needs
- Sharp cut in diesel and grid costs
- Monitored, maintained PV assets

Free fuel from the sun

2 Solar: Value Proposition

VALUE PROPOSITION

WHAT YOU GAIN	THE TAQA ARABIA EDGE
Lower Energy Bills Clean solar energy at lower cost than diesel-only pivot power, through long-term PPAs.	**Flexible Financing Solutions** CAPEX, BOOT/BOO or PPA — own it, transfer it over time, or buy cheaper solar under a zero-capex PPA.
Energy Security On-site generation reduces exposure to fuel-price spikes and supply disruption.	**Single Energy Partner** One single utility Provider for electricity and water and providing Both solar and Batteries for Maximum Savings and utilization
Sustainability & Carbon Cut Each 1 MWp avoids ~1,000 tons of CO_2 a year — strong ESG and export credentials.	**Lifecycle O&M & Guarantee** Remote monitoring, maintenance and performance guarantees keep output high for 25+ years.
Uses Idle Land Marginal and unused farmland becomes a productive, cost-saving energy asset.	**Egypt's Solar Pioneer** Operating Multiple solar Farms and water desalination projects across Egypt

Solar: Implementation Timeline

IMPLEMENTATION TIMELINE — START TO FINISH

Day 0–21
1
Site & Energy Study
Irradiation, pivot load & land survey.

Day 21–55
2
System Design
Array sizing & genset/battery-tie design.

Day 40–75
3
PPA Structuring
BOO / BOOT & tariff agreement.

Day 75–130
4
Procurement
Panels, inverters & mounting sourcing.

Day 130–210
5
Install & Commission
Build, tie-in, testing & go-live.

Day 210+
6
O&M
Monitoring & performance maintenance.

STUDY & DESIGN *Day 0–55*
Solar-irradiation and pivot-load study, land survey, array sizing and integration design with battery and genset systems.

FINANCE & BUILD *Day 40–210*
BOO/BOOT structuring and PPA tariff, procurement of panels, inverters and mounting, installation, tie-in and commissioning.

OPERATE *Day 210+*
Monitoring, cleaning and performance maintenance over the life of the PPA.

Solar: Proven Track Record

Dina Farms — Solar in Action

Powering one of Egypt's largest farms with renewable energy

TAQA Arabia operates renewable-energy projects at Dina Farms — one of Egypt's largest integrated agricultural operations — alongside its landmark Benban solar developments. It is direct proof that TAQA can deploy and run solar at true agricultural scale, powering irrigation and farm operations with clean energy.

Dina Farms
Live agricultural solar project

Benban
Landmark solar development

BOO / BOOT
PPA models, no farm capex

~1,000
t CO_2 avoided / MWp / yr

3 Battery Storage (BESS): Scope & How It Works

SCOPE

TAQA funds, builds, owns & operates the asset.
You pay only for the stored energy you use — little to no upfront CapEx.

1 TAQA INVESTS THE CAPEX

Battery energy-storage units · Power-conversion system & inverters · Switchgear & grid / solar interface · EMS & SCADA controls · BOO / BOOT financing

2 THE PROCESS — STEP BY STEP

1. TAQA sizes storage to the farm's load and solar profile.
2. Battery units and conversion systems are installed.
3. Storage charges from surplus daytime solar.
4. It discharges to run pumps and cold storage after dark.
5. An energy-management system optimizes every cycle.

3 WHAT YOU RECEIVE

- Round-the-clock power from daytime solar
- Backup through grid and supply gaps
- Stable energy for cold-chain and pumps

Solar after dark

Battery Storage (BESS): Value Proposition

TAQA Arabia

VALUE PROPOSITION

WHAT YOU GAIN

Solar After Sunset

Stores daytime solar surplus and dispatches it to run pivots into the evening and early morning.

Less Diesel, Lower Cost

Battery dispatch displaces costly diesel-genset hours, cutting fuel spend and engine wear.

Greener Every Year

More stored solar dispatched means less diesel and a steadily smaller carbon footprint.

A Stacked Asset

One system delivers solar-shifting, peak support and backup — value across multiple uses.

THE TAQA ARABIA EDGE

Flexible Financing Solutions

CAPEX, BOOT/BOO or PPA — own it, transfer it over time, or buy cheaper solar under a zero-capex PPA.

Single Energy Partner

One single utility Provider for electricity and water and providing Both solar and Batteries for Maximum Savings and utilization

Lifecycle O&M & Guarantee

Remote monitoring, maintenance and performance guarantees keep output high for 25+ years.

Smart Energy Management

TAQA Arabia EMS automatically optimizes charge/discharge against loads, solar and fuel cost to deliver the lowest Cost per Kwh or M3

Battery Storage (BESS): Implementation Timeline

IMPLEMENTATION TIMELINE — START TO FINISH

Day 0–21	Day 21–55	Day 40–75	Day 75–140	Day 140–220	Day 220+
1	2	3	4	5	6
Load & Solar Study	**System Design**	**Commercial Model**	**Procurement**	**Install & Commission**	**Smart O&M**
Pivot profile & solar-surplus analysis.	BESS sizing & hybrid configuration.	BOO / BOOT & savings agreement.	Batteries, PCS & EMS sourcing.	Build, integration, testing & go-live.	EMS optimisation & maintenance.

STUDY & DESIGN *Day 0–55*

Pivot-load and solar-surplus analysis, BESS sizing and hybrid configuration with solar and gensets for 24-hour supply.

FINANCE & BUILD *Day 40–220*

BOO/BOOT structuring and savings agreement, procurement of batteries, PCS and EMS, installation, integration and commissioning.

SMART OPERATE *Day 220+*

EMS-driven charge/discharge optimisation against loads, solar and fuel cost, plus monitoring and maintenance.

Battery Storage (BESS): Proven Track Record

Solar-Plus-Storage Integration

Turning intermittent solar into round-the-clock farm power

As Egypt's largest private power player, TAQA Arabia integrates Battery Energy Storage with its solar, diesel and gas-genset assets to deliver dispatchable, lower-carbon energy. For the farm, storage is the piece that lets solar run the pivots long after the sun goes down — the heart of the integrated power stack.

Solar shifting
Day energy used at night

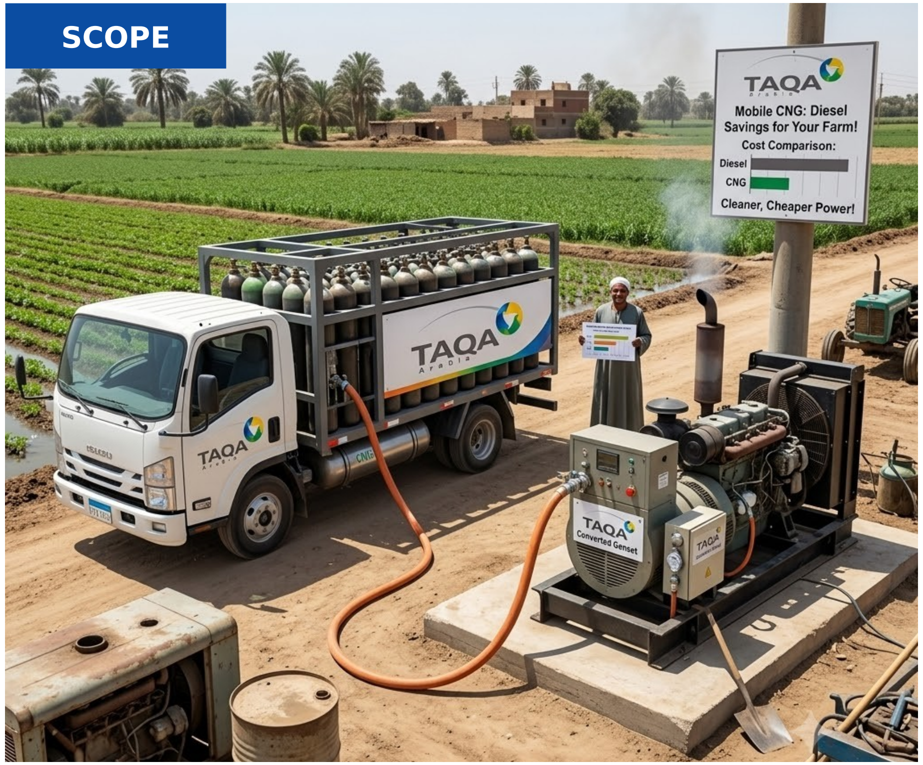
Peak support
Firms the power stack

BOO / BOOT
No farm capex

24/7
EMS monitoring & O&M

Mobile CNG: Scope & How It Works

SCOPE



TAQA funds, builds, owns & operates the asset.

You pay only for the gas you consume — little to no upfront CapEx.

1 TAQA INVESTS THE CAPEX

Mother station & compression · CNG / virtual-pipeline trailers · On-site decompression & PRMS skid · Metering & safety controls · BOO / BOOT financing

2 THE PROCESS — STEP BY STEP

1. Gas is compressed at a TAQA mother station to ~250 bar.
2. Trailers deliver it to the farm as a virtual pipeline.
3. On-site skids decompress and regulate to process pressure.
4. Metered gas fuels dryers, greenhouses and processing.
5. TAQA tracks usage and refills ahead of demand.

3 WHAT YOU RECEIVE

- Pipeline-grade gas without a pipeline
- Lower-cost fuel for drying & processing
- Managed supply and refills

Gas, off-grid

Mobile CNG: Value Proposition

VALUE PROPOSITION

WHAT YOU GAIN

Off-Grid Gas Supply

Reliable natural gas to remote farms with no pipeline access, by mobile virtual pipeline.

Cost Savings vs. Diesel

≈40% lower fuel cost than diesel — a major cut to the farm's biggest running expense.

Cleaner Operations

~24% lower CO_2 than diesel helps the farm meet tightening emissions and export requirements.

99.5% Uptime SLA

SCADA-monitored hot-swap replenishment guarantees uninterrupted gas for the gensets.

THE TAQA ARABIA EDGE

Master Gas Scale and Network

TAQA's Master Gas runs Egypt's leading CNG virtual pipeline and has CNG stations scattered Across Egypt

Flexible Financing Solutions

CAPEX, BOOT/BOO or PPA — own it, transfer it over time, or buy cheaper solar under a zero-capex PPA.

Scalable to Any Load

Starter (500 Nm^3/day) to Heavy (5,000+ Nm^3/day) — scales with the farm's gas demand.

Lifecycle O&M & Guarantee

Remote monitoring, maintenance and performance guarantees keep output high for 25+ years.

Mobile CNG: Implementation Timeline

IMPLEMENTATION TIMELINE — START TO FINISH

Day 0	Day 1–3	Day 4–7	Day 8–14	Day 15–30	Day 90–120	Day 120–200
1	2	3	4	5	6	7
Discovery	**Initial Contact**	**Site Audit**	**Contract**	**Mobilization**	**On-Site Install**	**Live Gas**
Lead identification & qualification.	NDA signed, requirement scoping.	Gas load profiling & demand curve.	Capacity tier, CaaS pricing & 99.5% SLA.	MEGC & PRU procured and prepared.	PRU adapter installed, SCADA activated.	Switchover from diesel to live CNG supply.

DISCOVERY & AUDIT *Day 0–14*

Lead qualification, NDA, site visit and full gas-load profiling — output is a precise demand curve and connection plan.

CONTRACTING *Day 8–14*

Capacity tier selection (Starter / Industrial / Heavy), CaaS pricing model, 99.5% uptime SLA and payment terms.

DEPLOYMENT & GO-LIVE *Day 90–200*

MEGC mobilized, PRU adapter installed, SCADA activated, safety tests and switchover from diesel to live CNG supply.

Mobile CNG: Proven Track Record

Virtual Pipeline — 4 Governorates

First company in Egypt to supply natural gas through a mobile virtual pipeline

TAQA Arabia pioneered mobile CNG in Egypt, using its network of 86 CNG stations to extend a virtual pipeline into four governorates with no fixed gas infrastructure. The same model brings clean, lower-cost gas to off-grid farms and agribusiness — fueling gensets that would otherwise burn diesel.

86
CNG stations feeding the virtual pipeline

4
Governorates served off-grid

+2,350
mmscf CNG delivered per year

+10
Active mobile-CNG clients

Why One Partner: The TAQA One-Stop-Shop

From Five Vendors to One Operator

One SLA

A single agreement covers water, solar, storage, gas and diesel — one uptime guarantee for the whole farm.

One Communication Point

A single account team and 24/7 hotline — no chasing separate water and power contractors.

One Commercial Relationship

Consolidated billing and aligned BOO/BOOT terms instead of separate procurement cycles.

One Smart System

Water and the full power stack run on one TAQA energy-management platform — optimised together.

One Accountable Owner

End-to-end responsibility removes interface risk — TAQA owns the farm's water-and-energy uptime.

The Integrated Economics: Diesel-Only vs. TAQA

What changes when one partner runs an optimised water-and-power stack — illustrative

TODAY — DIESEL-ONLY FARM

High, volatile fuel bill
Diesel is the biggest farm cost — exposed to every price spike.

Pivots stop on failure
Stand-alone gensets fail with no backup — lost irrigation cycles.

High carbon footprint
100% diesel means maximum CO_2 and emissions per feddan.

Multiple vendors
Separate water, fuel and genset suppliers — no accountability.

WITH TAQA — INTEGRATED STACK

Up to ~40% lower energy cost
Solar + storage + CNG displace most diesel hours — predictable PPA tariffs.

24/7 irrigation uptime
Batteries and multi-source firming keep pivots turning through any gap.

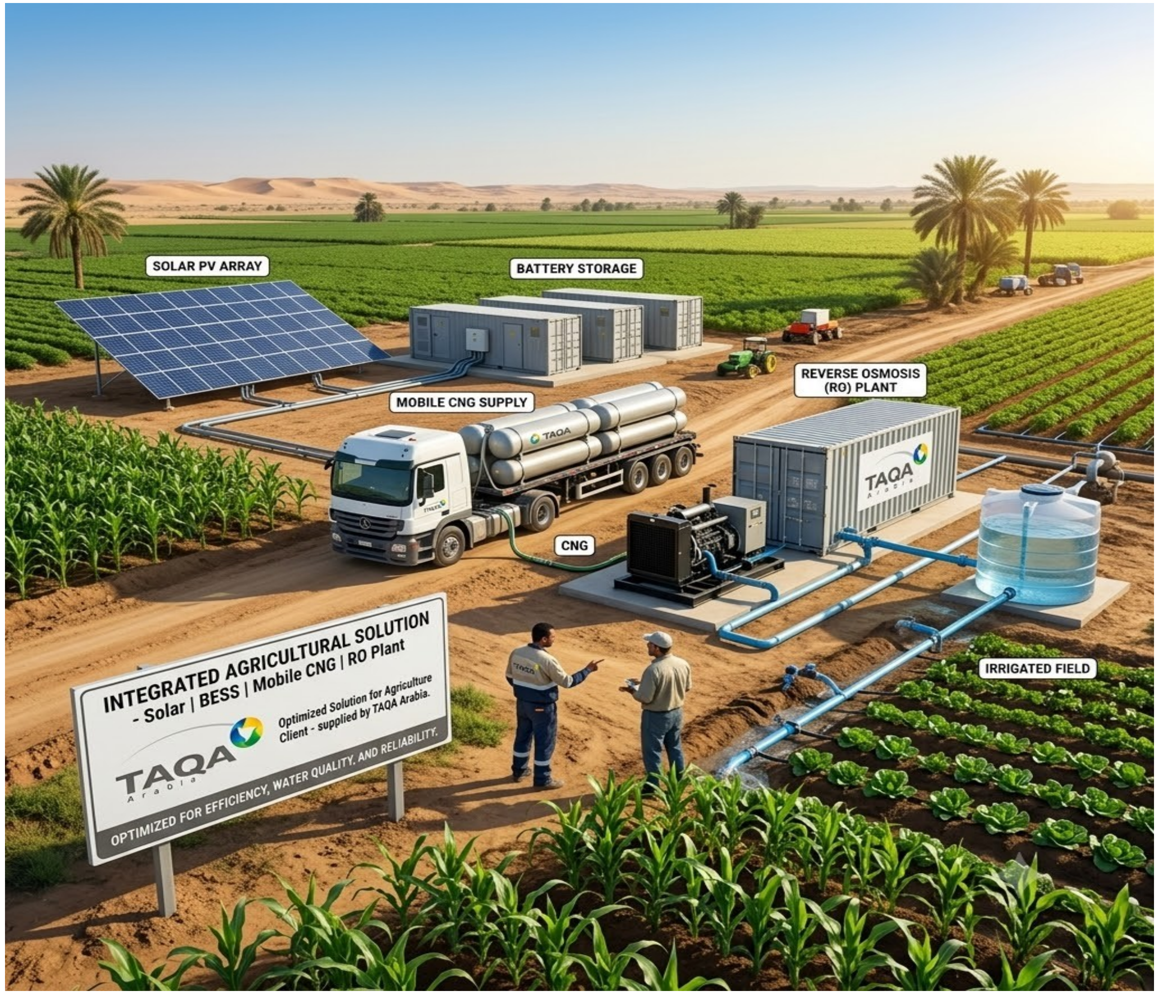
Sharply lower CO_2
Each 1 MWp of solar avoids ~1,000 t CO_2/yr; CNG cuts a further ~24% vs diesel.

One accountable partner
Single SLA for water + power, zero capex under BOO/BOOT, one smart system.

The result: lower and more predictable energy cost, secured water, round-the-clock irrigation and a stronger ESG story — from one partner, with no capex.

Figures are illustrative / market based and depend on farm size, crop, irrigation schedule and fuel prices.

Proven on the Ground: Dina Farms & Beyond



Renewable energy at one of Egypt's largest farms

TAQA Arabia operates renewable-energy projects at Dina Farms — one of Egypt's largest integrated agricultural operations — alongside its landmark Benban solar developments. These projects prove TAQA can deploy and run clean energy at true agricultural scale. The next step is the fully integrated stack: pairing that solar with storage, gas and water under one operator.

What we bring to your farm:

- Water Solutions
- Solar
- Battery Storage
- Mobile CNG

Dina Farms
Live agricultural solar

Benban
Landmark solar project

~1,000
t CO_2 / MWp / yr avoided

Zero
Capex under BOO/BOOT